Wired 02 | Provisioning a Campus Fabric

Overview

In this lab, you'll onboard your switch using CloudVision Studios, adding a new campus leaf switch to an existing Campus Fabric. We'll take a switch from out-of-the-box Zero-Touch Provisioning (ZTP) mode to a fully provisioned member of the fabric — without the use of CLI or a console cable.

What you'll learn:

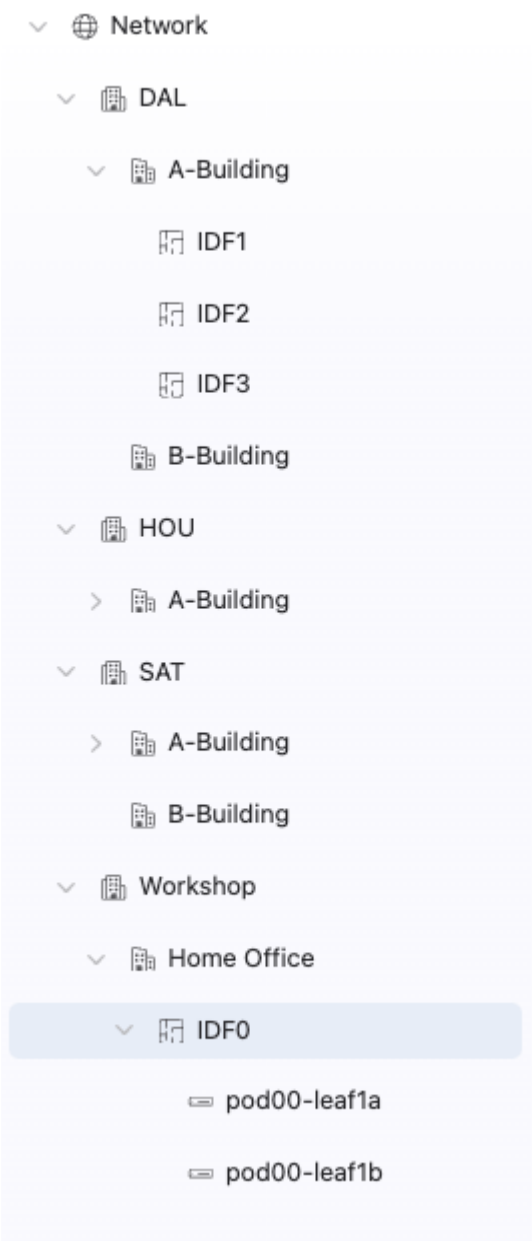
- Onboard a new switch using CloudVision Studios Quick Actions
- Navigate the Campus Fabric hierarchy (Campus, Pod, Access Pod)
- Configure spine uplinks for external devices
- Review and submit a workspace for change control
- Execute and monitor a change control

Your environment has been pre-configured with a sample Campus to assist with these new concepts. Studios uses flexible constructs to describe your campus footprint:

- **Campus Fabric**: Your campus `Workshop` has been created — this is equivalent to a site or region.
- **Campus Pod**: Your pod `Home Office` could be a building or MDF.
- **Access Pod**: Your access pods `IDF1` and `IDF2` could be a floor or IDF.

Show me an example!

CloudVision Network Hierarchy gives a better visual representation of how this might look in a real environment.



Out with the console cable

Zero-Touch Provisioning in this lab is performed using a bootstrap script delivered over DHCP option 67. This tells your device where to begin streaming for onboarding. You can read more on the CloudVision Help Center.

Open Launchpad | **Login Help** | **Lab Topology**

Onboarding Devices

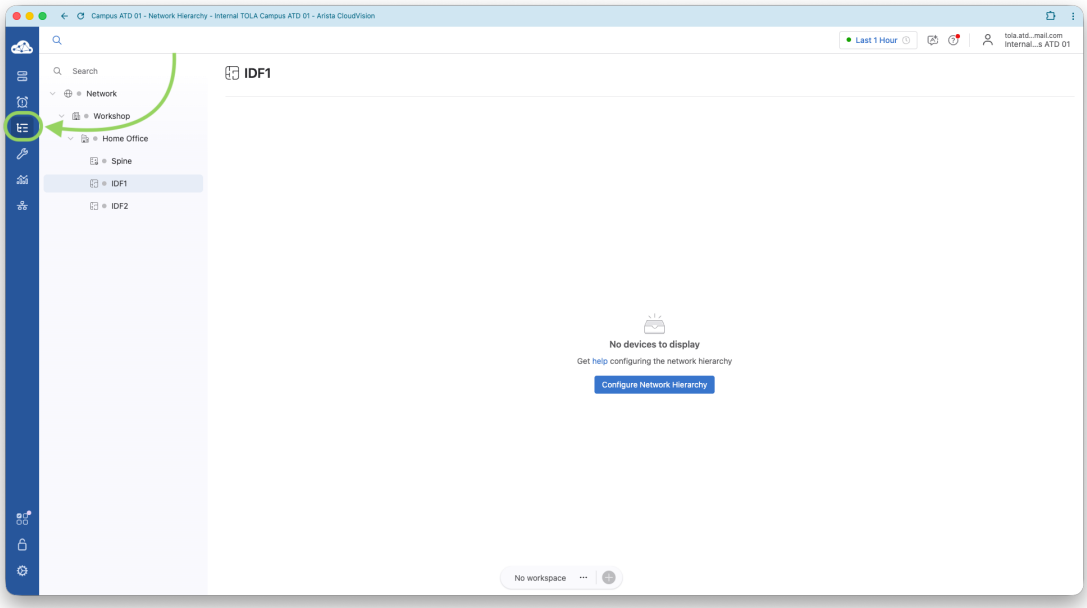
1. Before you begin, we've assigned you and a fellow student to a `Pod`, and each `Pod` is assigned to a separate CloudVision Organization. TOLA-ATD-(01-12). We are using a bootstrap server to distribute tokens per VLAN which are assigned to each Organization.

   IDF Assignments

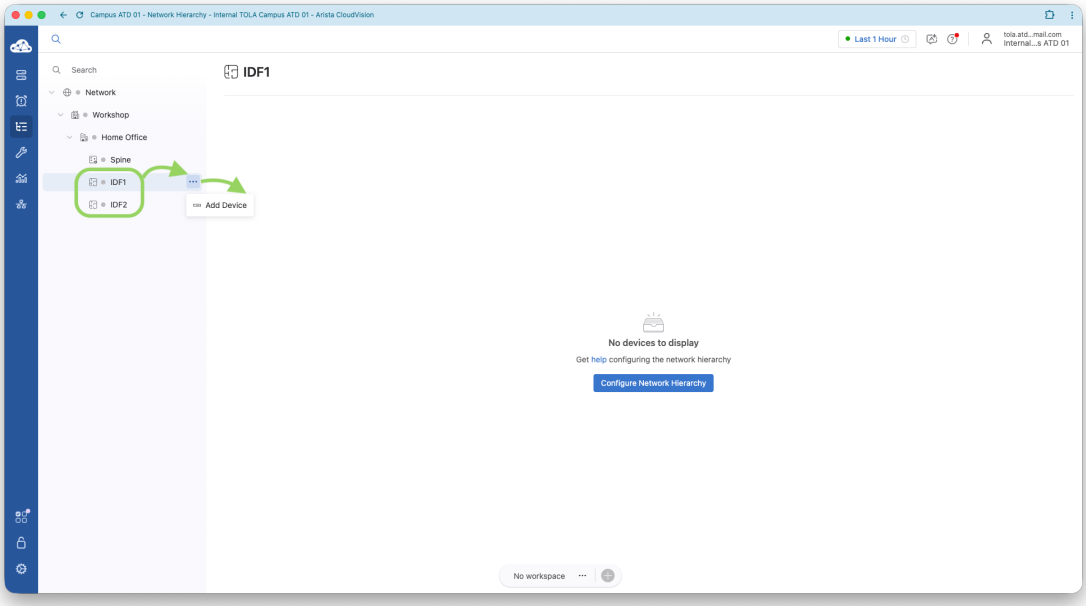
   You and and a fellow student will work separately to onboard your switches to you assigned IDFs.

   Student-1 is assigned to IDF1 and Student-2 is assigned to IDF2.

2. To begin onboarding your switches; Navigate to the `Network Hierarchy` tab on the left if you are not already there.



3. Under Network, Workshop, and Home Office, hover your mouse to the right of IDF1 or IDF2. Select the three-dot menu and select `Add Device`.



4. This workflow will take you through onboarding the device. Use the table below determine the `Hostname` of your device.

What's my hostname?

Student1 which choose LEAF1A and Student2 which choose LEAF1B for each Pod

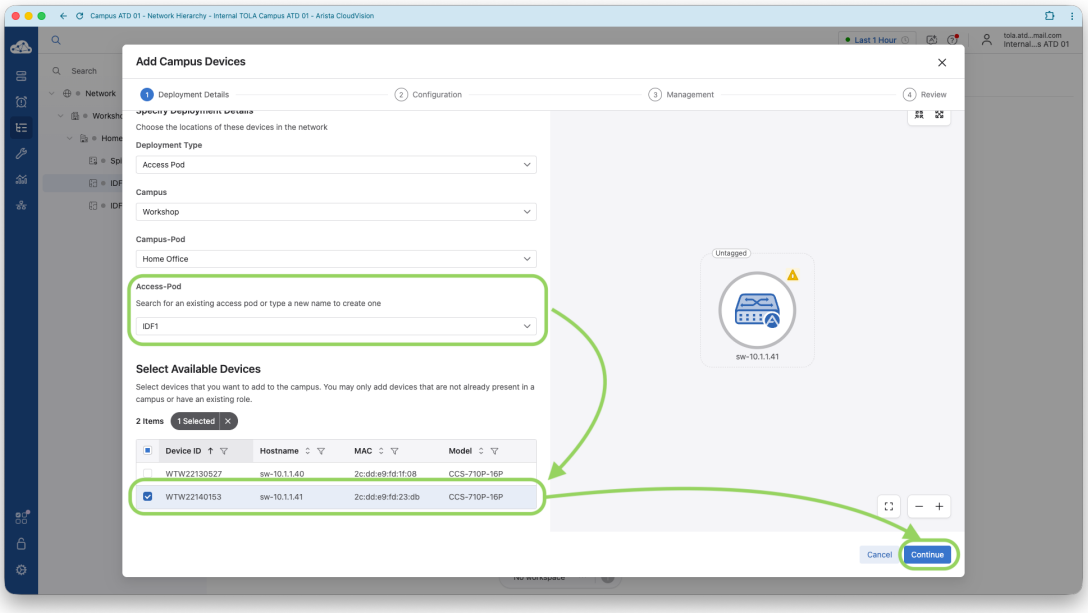
Switch Names

Switch Name	Serial
POD01-LEAF1A	WTW22140153
POD01-LEAF1B	WTW22130527
POD02-LEAF1A	WTW22130402
POD02-LEAF1B	WTW22150107
POD03-LEAF1A	WTW22130409
POD03-LEAF1B	WTW22130472
POD04-LEAF1A	WTW22130476
POD04-LEAF1B	WTW22130478
POD05-LEAF1A	WTW22130506
POD05-LEAF1B	WTW22140213
POD06-LEAF1A	WTW22140174
POD06-LEAF1B	WTW22130505
POD07-LEAF1A	WTW22130459
POD07-LEAF1B	WTW22140623
POD08-LEAF1A	WTW22130479
POD08-LEAF1B	WTW22130554
POD09-LEAF1A	WTW22140124

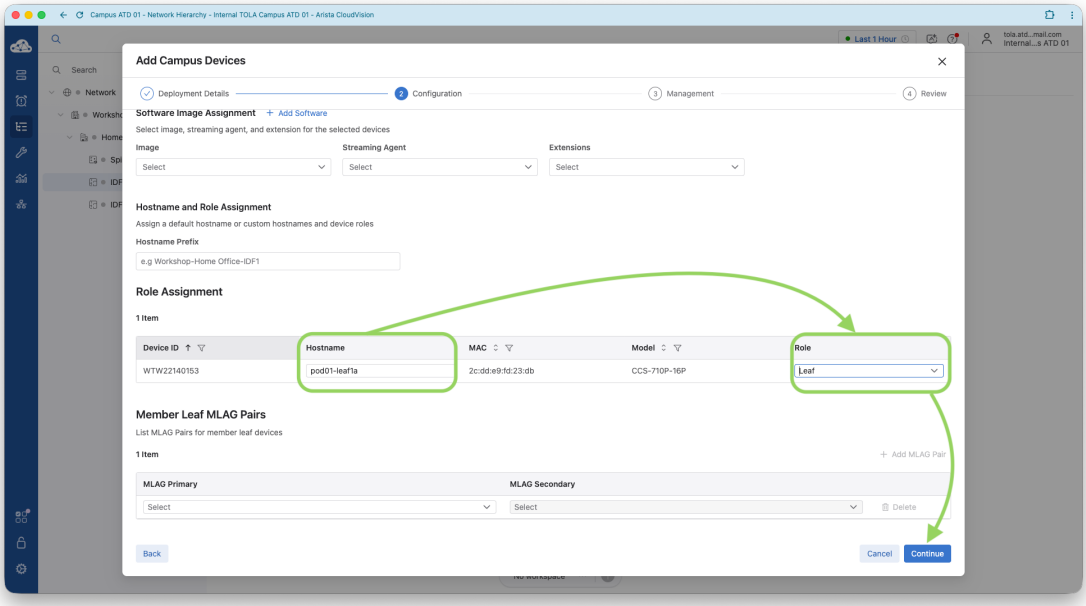
Switch Name	Serial
POD09-LEAF1B	WTW22130439
POD10-LEAF1A	WTW22130436
POD10-LEAF1B	WTW22130545
POD11-LEAF1A	WTW22130549
POD11-LEAF1B	WTW22130529
POD12-LEAF1A	WTW22130440
POD12-LEAF1B	WTW22130532

Step 1

Select your switch via the serial number then select `Continue`.



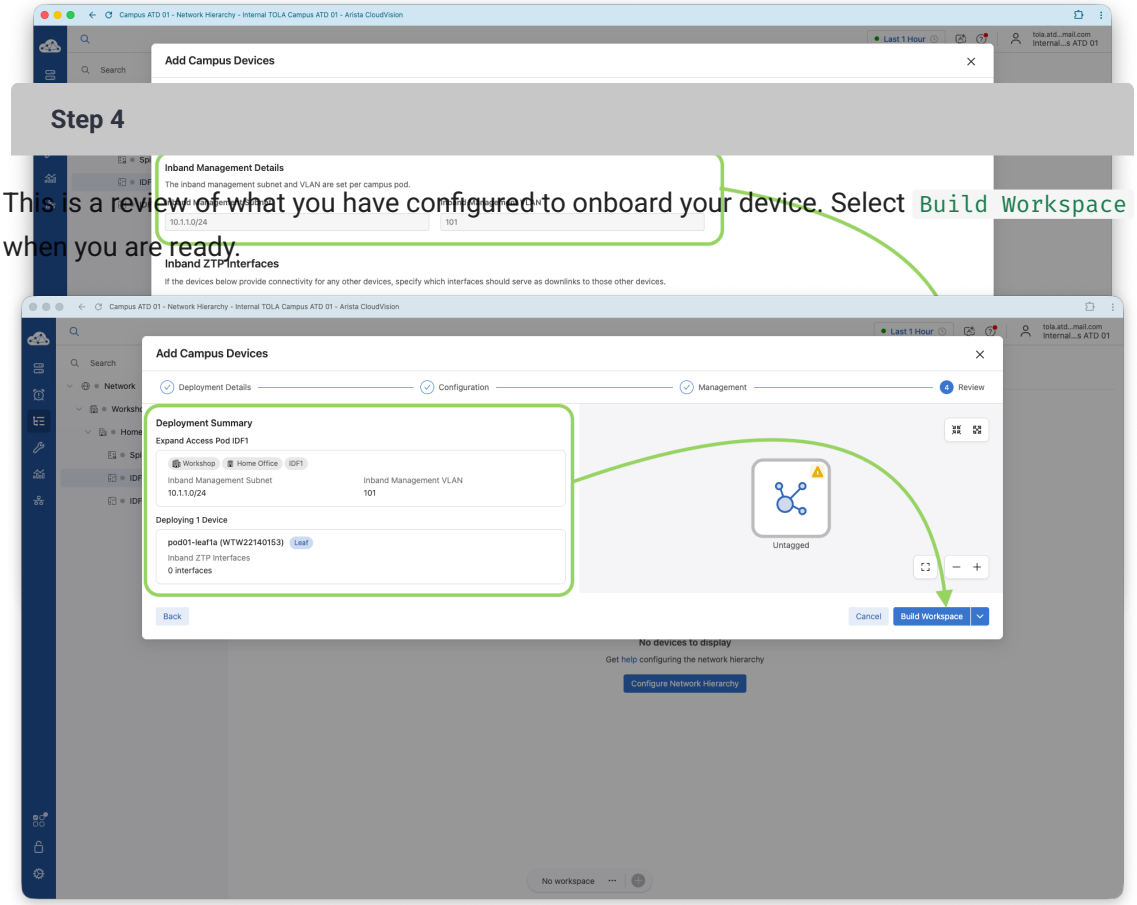
In `Role Assignment`, select the `Hostname` field and name your switch `pod##-leaf1X`, where `##` is your pod number and `X` is either student A or B. Select the role as `Leaf` and select `Continue`.



Step 3

Validate your `Inband Management Subnet` and `Inband Management VLAN` match your assigned pod number. This was pre-configured in Studios.

- **Subnet**: `10.1.#.0/24` where `#` is the pod number
- **VLAN**: `1##` where `#` is the pod number

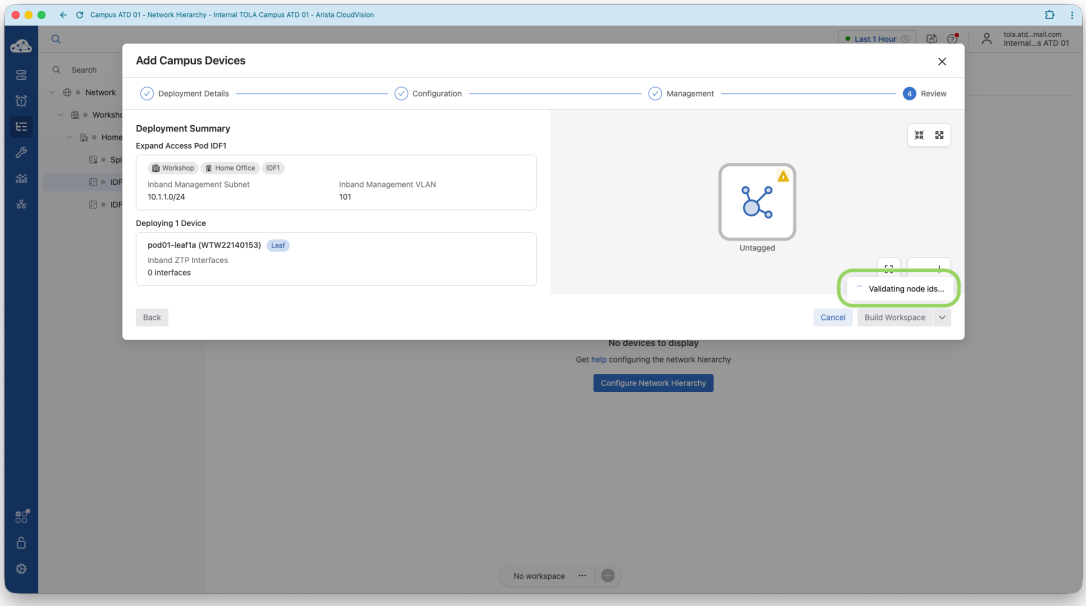


Step 4

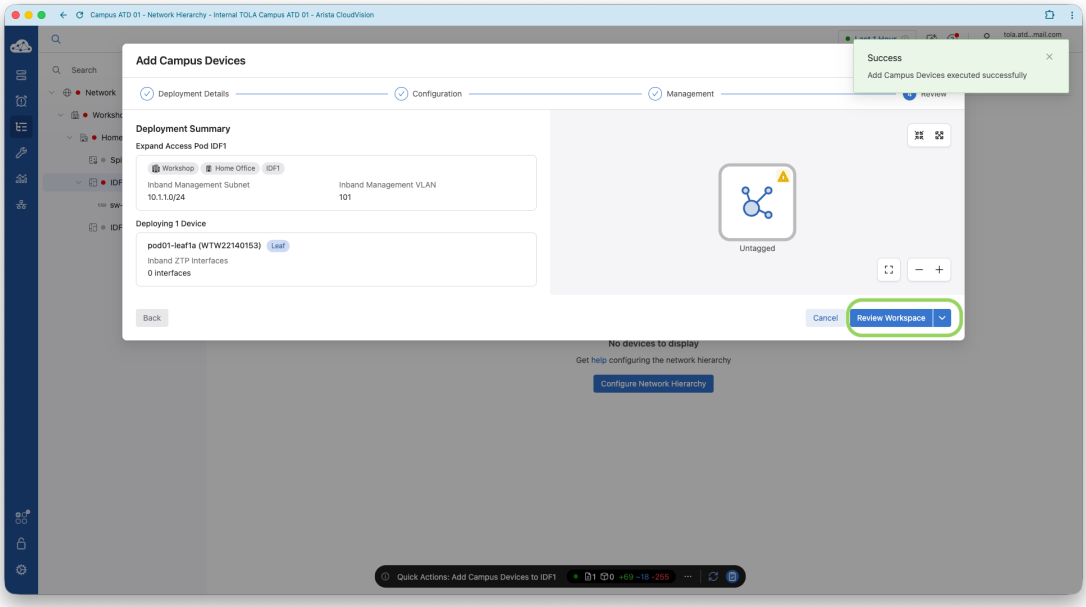
This is a review of what you have configured to onboard your device. Select `Build Workspace` when you are ready.

Step 5

During this step, the Quick Action is adding your devices to the Studios inventory and generating configuration for your new device. If we had selected an EOS image, a task would have been created to upgrade the device during onboarding.



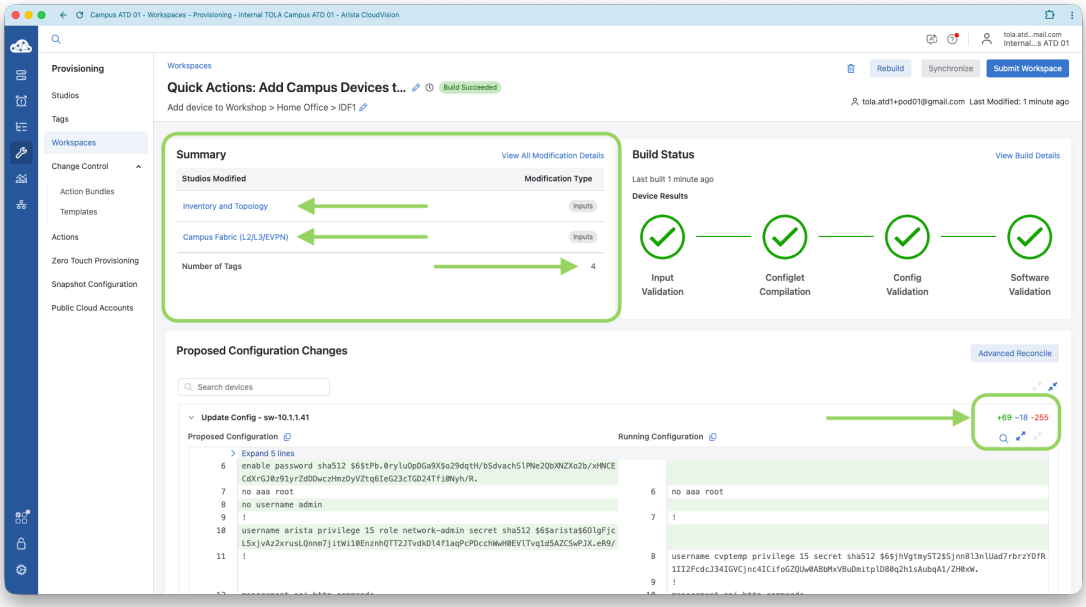
5. The workflow is completing several steps to onboard the device. Select `Review Workspace` to explore.



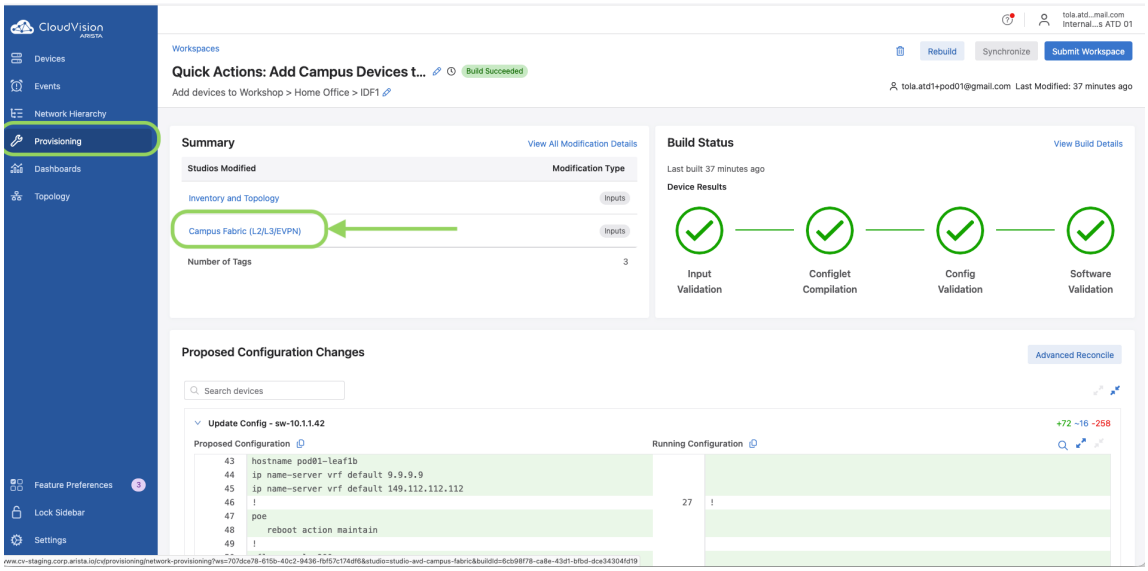
6. In the workspace, note the top-level `Summary` box. There are several studios modified:
    a. `Inventory and Topology`: Devices selected are added to the Campus inventory.
    b. `Campus Fabric (L2/L3/EVPN)`: Devices were added to their respective Campus Access Pod `IDF1` and will inherit configuration from that part of the campus.

c. Note the number of Tags

d. Review the number of lines being added (green), modified (blue), and deleted (red) in the bottom right.



7. Do **not** select `Submit Workspace`. Navigate back to the `Studios` by selecting `Campus Fabric (L2/L3/EVPN)` or going back to 🔧 `Provisioning`.



Did you Submit your WorkSpace

It's okay since No Configuration change has been made. You will need to approve a Change Control later to configure the switch. Also, a new workspace will be automatically created when you begin making your next change.

Configuring Spine Uplinks

1. Configuring the Campus Fabric `Campus Fabric (L2/L3/EVPN)`.

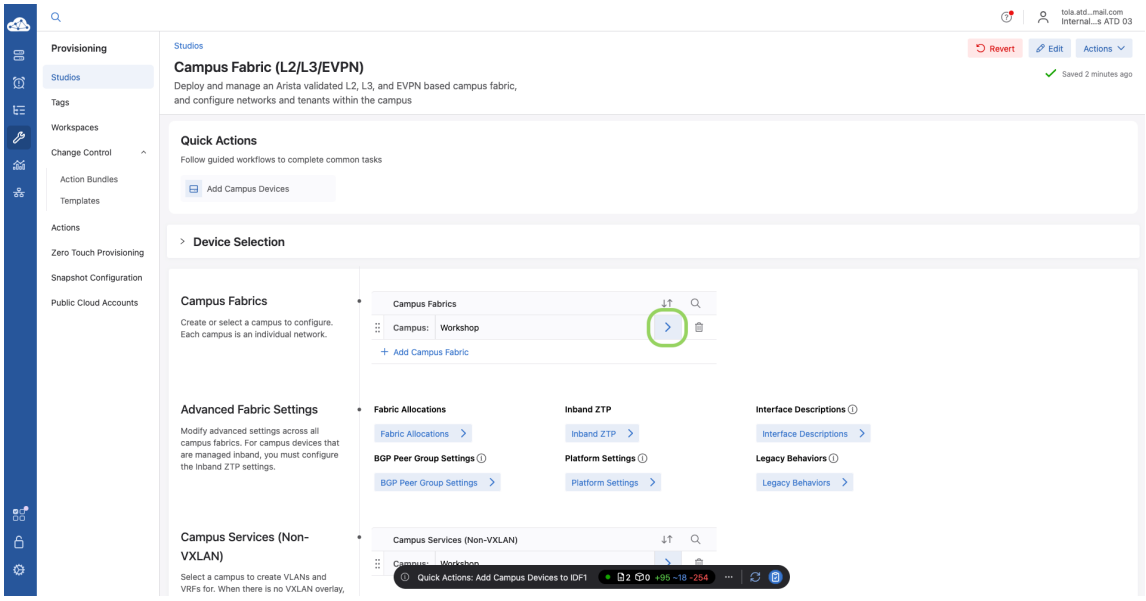
Spine Configuration

Normally, you wouldn't have to perform this step; however, since the spines are part of the Instructor's Fabric, we will need to manually configure the spine uplinks.

2. Next, we'll drill down to configure the spine uplinks.

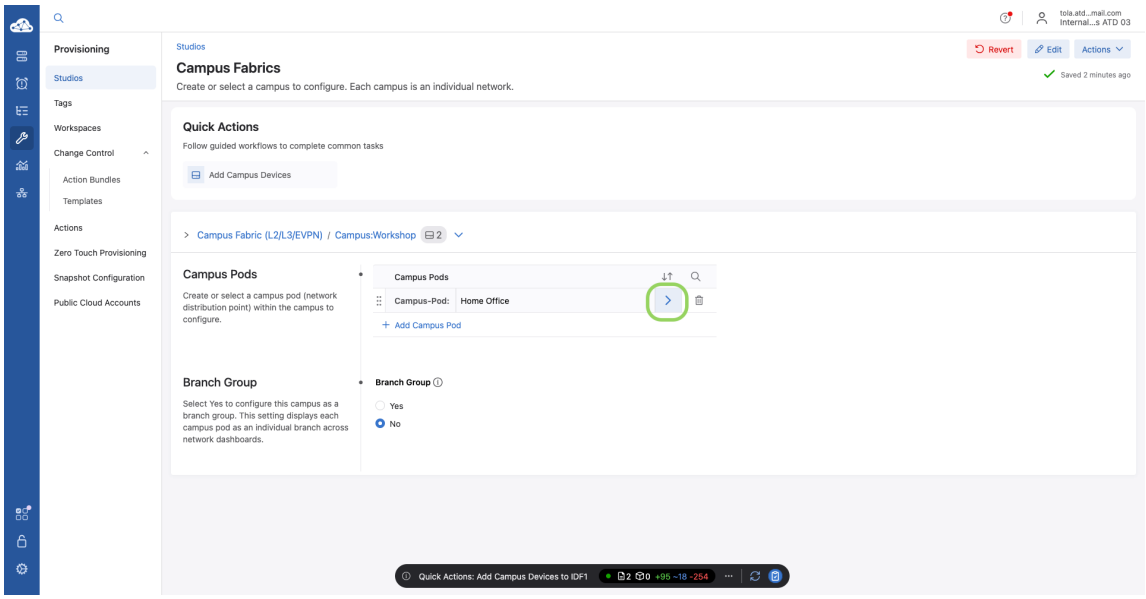
Step 1

Select the blue arrow by `Campus: Workshop`



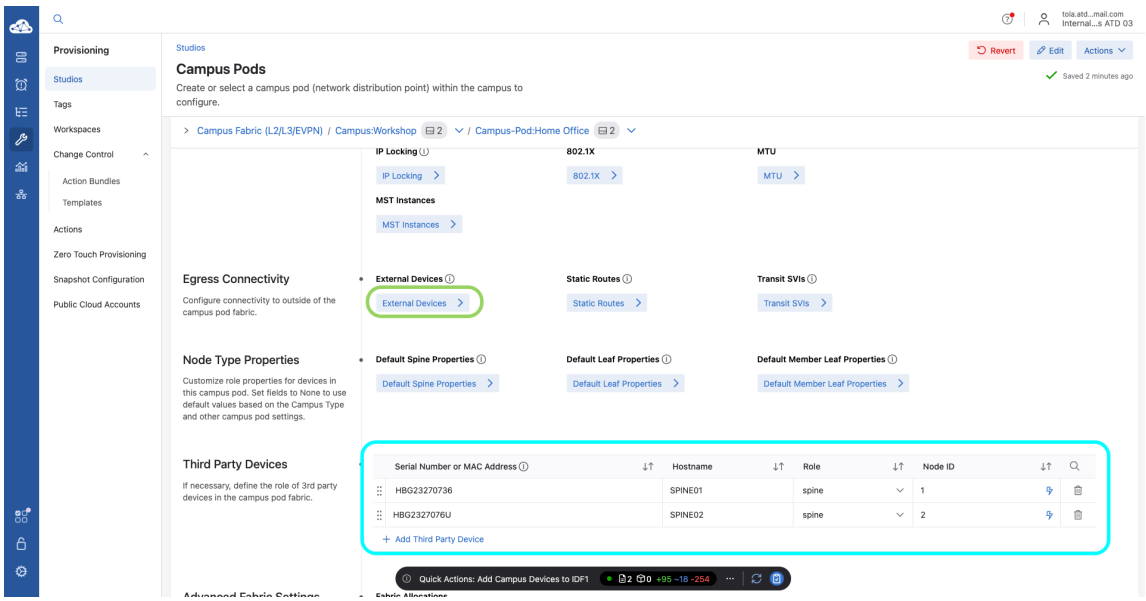
Step 2

Select `CampusPod: Home Office`



Step 3

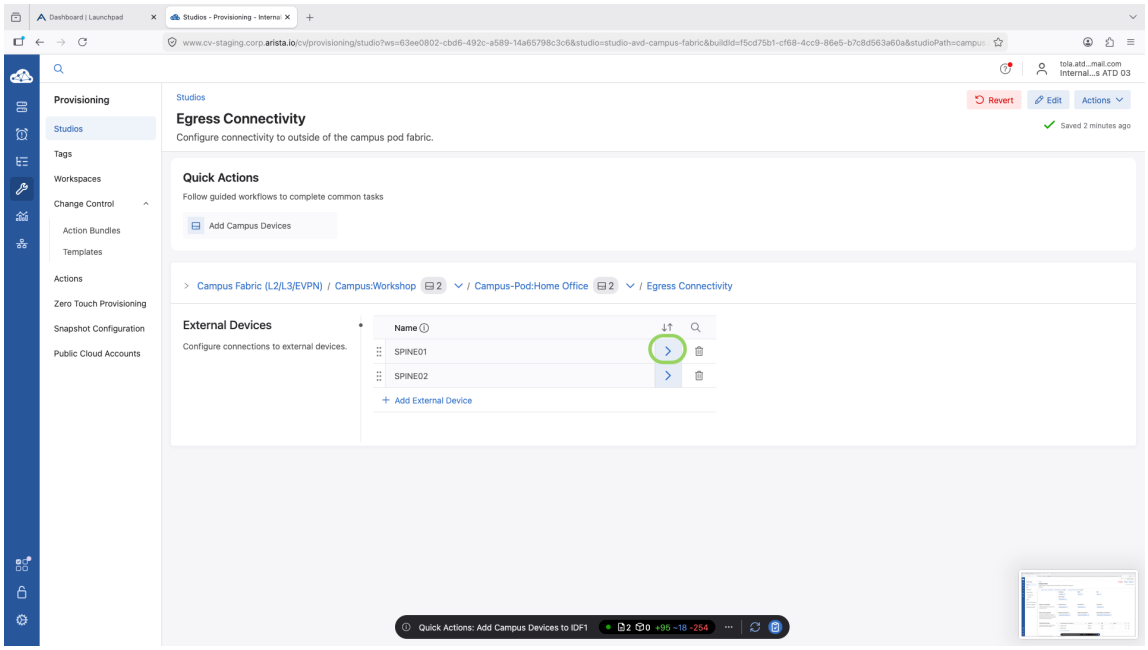
Scroll down to `Egress Connectivity` and select `External Devices`. Note the spines are configured as 3rd party devices below.



3. Now let's configure your switch uplinks for both Spines. Each switch has uplinks to both SPINE01 & SPINE02 on Ethernet Interfaces 15 & 16.

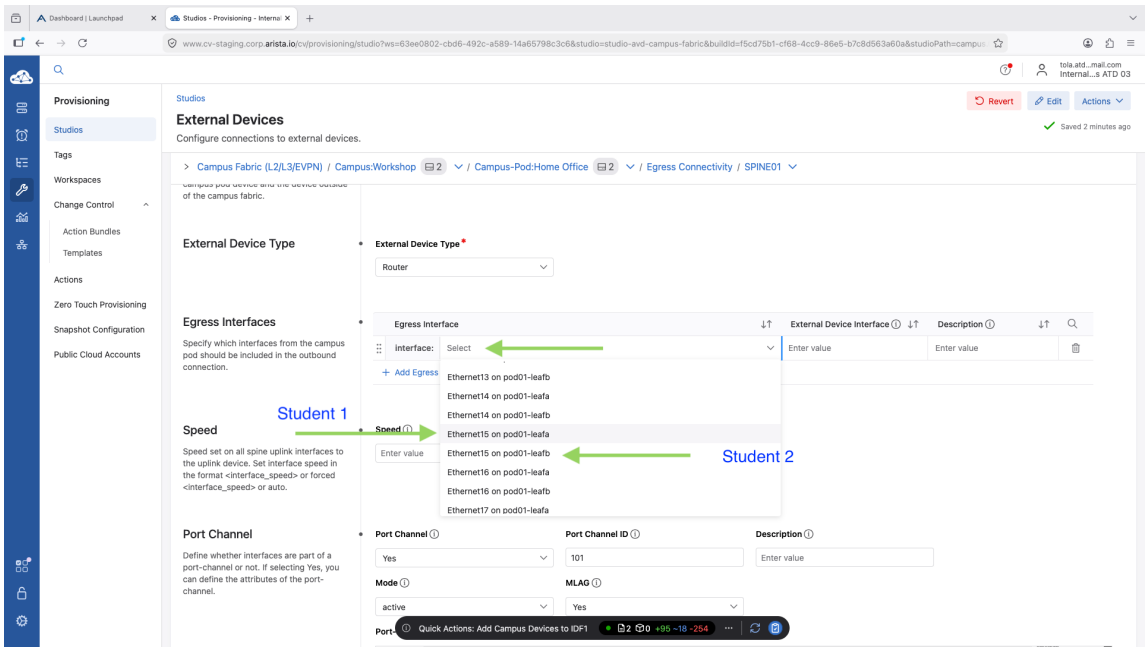
Step 1 - Select SPINE01

Select the blue arrow next to SPINE01.

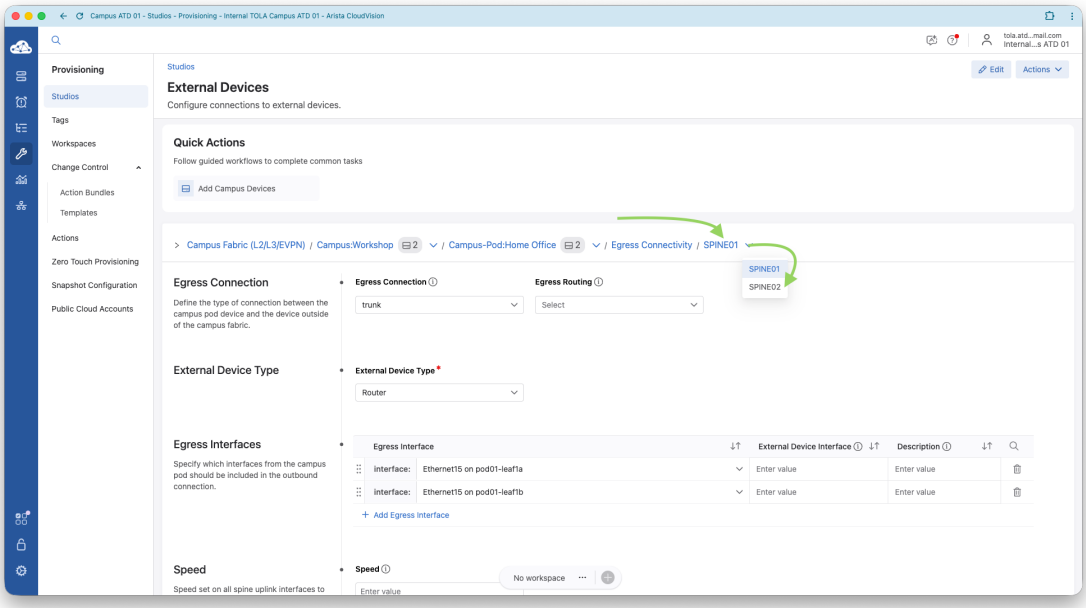


Step 2 - Configure SPINE01

Scroll down to `Egress Interfaces`, then select the blank space next to `Interface` and type `15` in the `Interface:` and select `Ethernet 15 on podXX-leaf(a for student 1 and b for student 2)`.

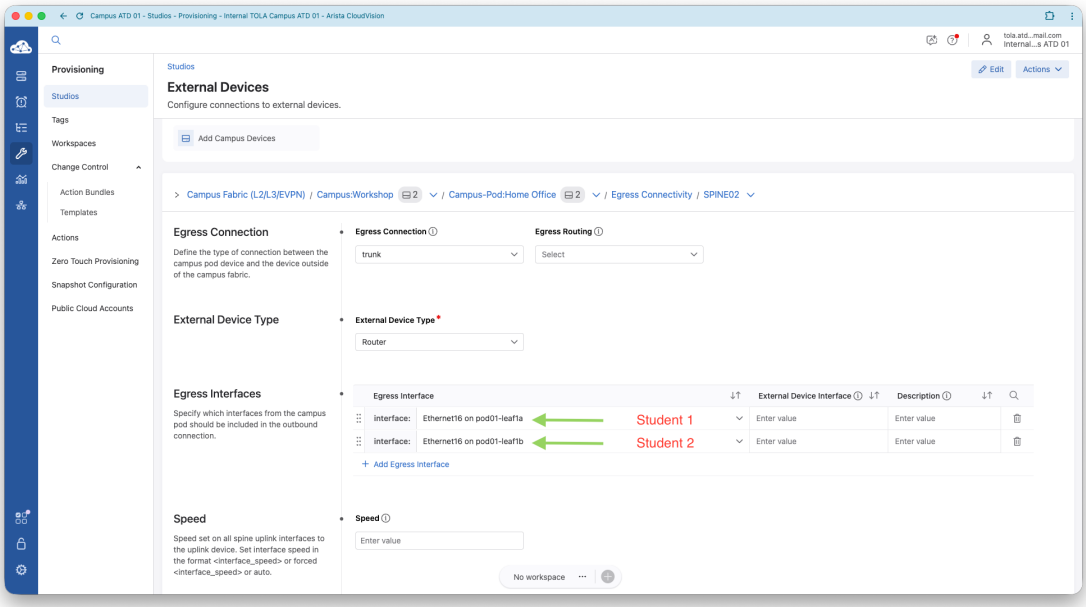


Select the blue down arrow next to SPINE01 to collapse it, then expand SPINE02.



Step 4 - Configure SPINE02

Do the same for the interface 16, selecting `Ethernet 16 on podXX-leaf(a for student 1 and b for student 2)`.



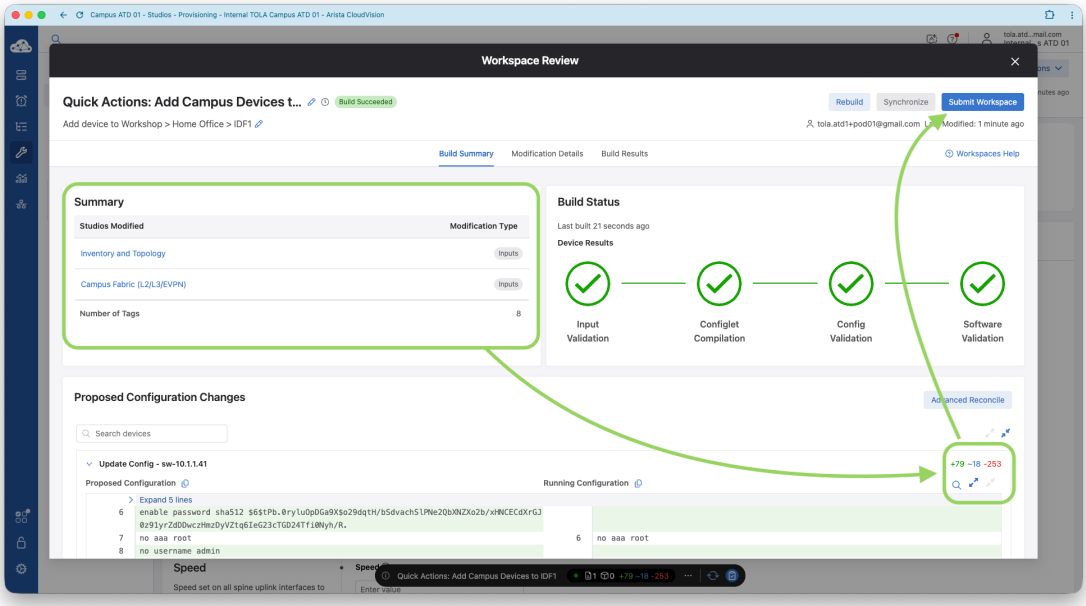
4. Let's now review our workspace and the changes we've made. Select `Review Workspace` on

the Workspace Island.

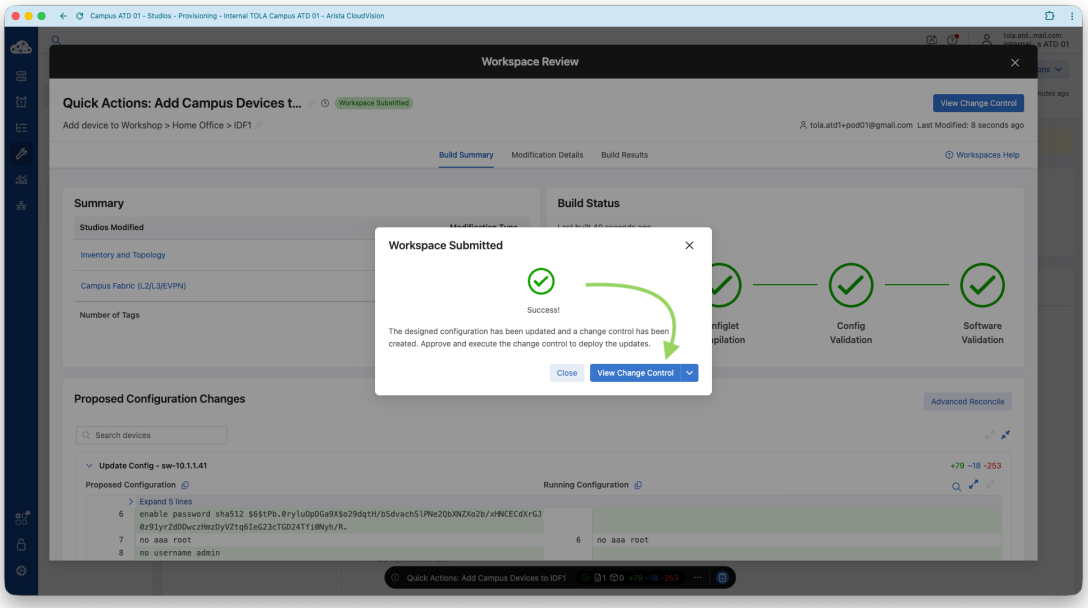


5. You can review the configuration changes to the devices.

> **The Summary**
>
> You can see there are 2 Studios and 8 Tags being used. We've also pre-configured several other Studios (Authentication, DNS, Date/Time, Streaming Telemetry) for the base configuration of the switch instead of using a Static configuration.



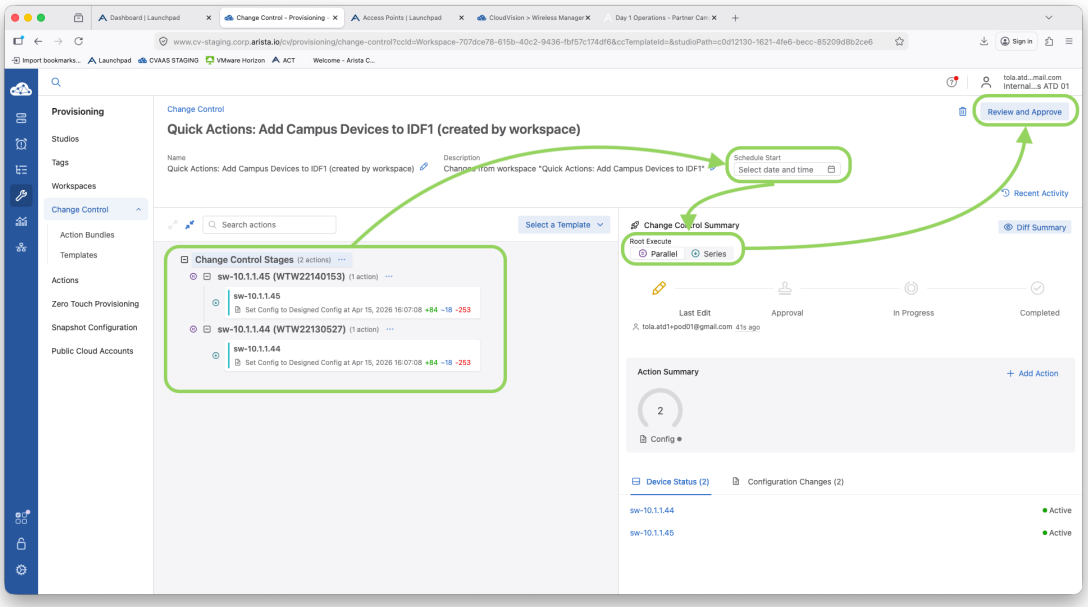
6. Select `Submit Workspace` to generate the Change Control.



7. Select `View Change Control` — let's explore executing a change control in the next section.

Executing a Change Control

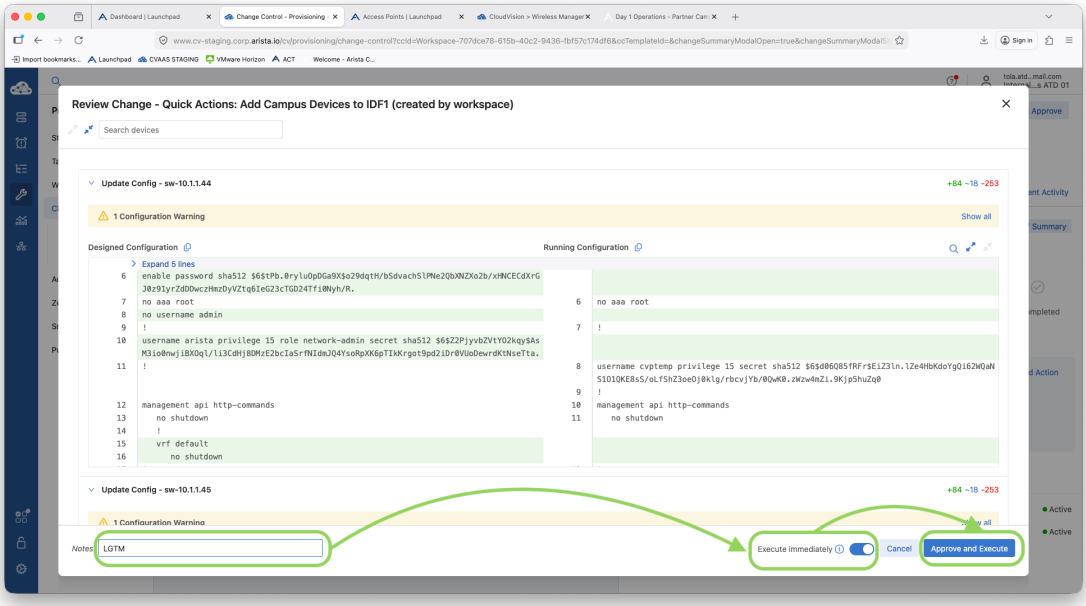
1. Within the Change Control, you can review the configurations as we just did in the workspace. This encourages review of changes prior to execution.



2. When you are ready, select `Review and Approve` at the top right, toggle `Execute`

`Immediately`, and select `Approve and Execute`.

> **I approved my own change?!**
>
> Yes, for this workshop you can approve your own change. In `Settings > General Settings`, the toggle for `Non-Author Change Control Review` can be enabled to enforce a change control review from a "non-author".



3. Let's explore what is going on during execution of the change control. While this is happening, feel free to select `Logs` near the top right to watch what's happening.

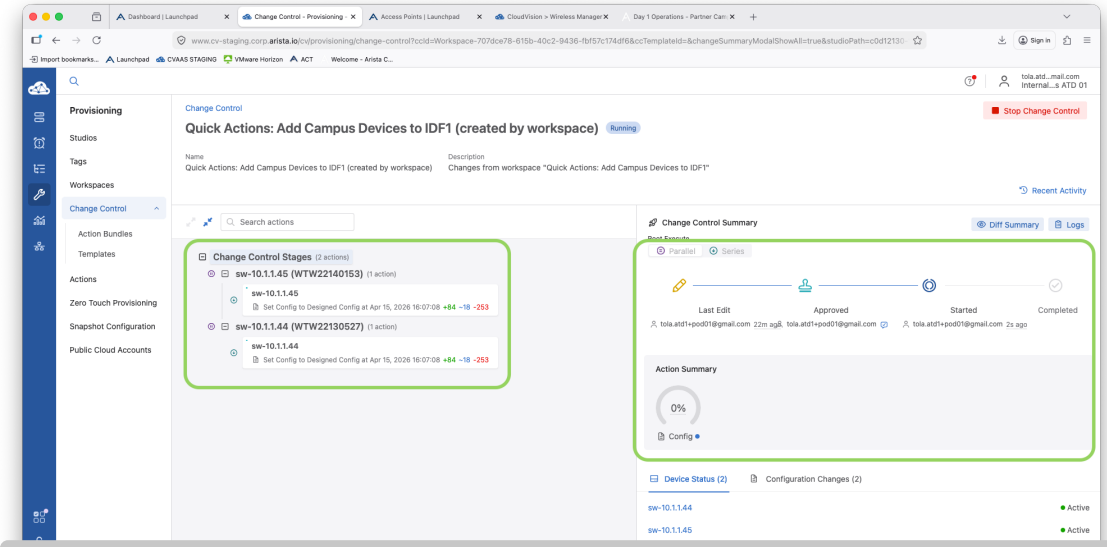
CC Start

So what is the device doing?! There are a number of things the device goes through during a ZTP onboard:

a. The device configuration will be applied to the device.

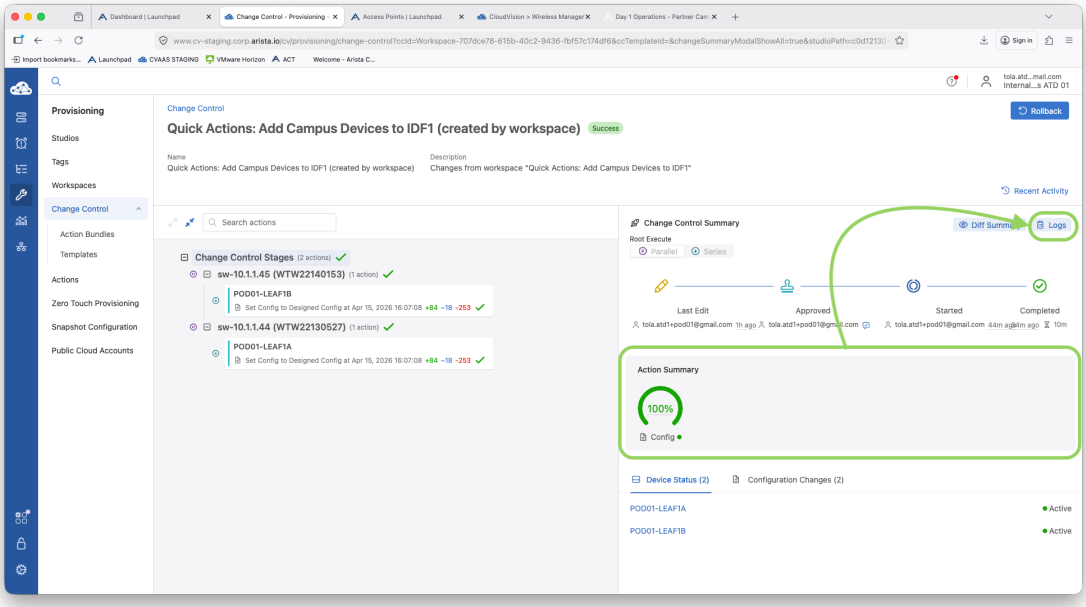
b. The switch will reboot to bring the device out of ZTP mode (this will take about 10 minutes).

c. We did not upgrade the device, but if we had, the software upgrade would then take place and reboot once more.



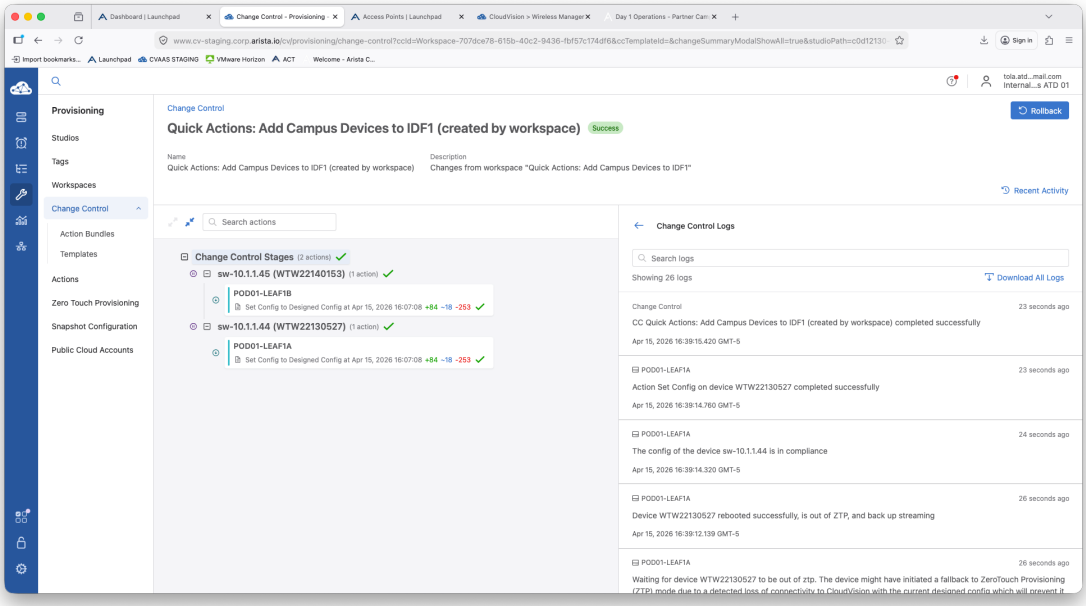
CC Finish

Now that the change control has finished, you will see a summary of any tasks that failed.

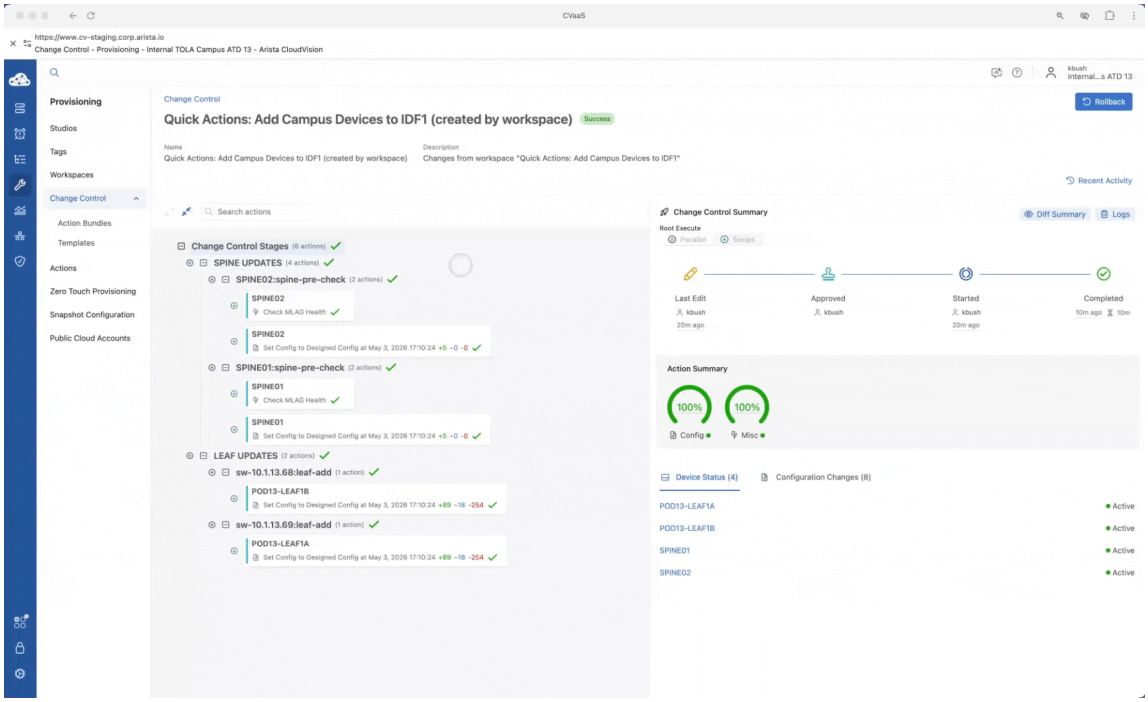


CC Device Logs

There are a number of logs you can review. The list is long, but provides great detail about what's going on behind the scenes.



4. You can now navigate back to the `Network Hierarchy` tab on the left and see your devices streaming with their new hostname!



5. That's it! Your new campus switch went from out-of-the-box ZTP mode to a configured member of the Campus Fabric. We're going to explore further changes to this switch in the next lab!

🎉 CONGRATS! You have completed this lab! 🎉

LET'S GO TO THE NEXT LAB!

2026-05-20